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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
First Semester M.Tech.(CCT)(CBCS) Degree Examinations
 January 2023

22CCT123 - PAVEMENT DESIGN AND CONSTRUCTION

Duration: 3 Hours

Max. Marks:100

- Note:** 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.
 2) Assume all the missing data suitably as per IRC.
 3) Use of Provided copy of IRC Design Charts and Tables is permitted.

Unit – I

Marks BT* CO* PO*

1. a) The load - penetration values of CBR tests conducted on two soil specimens of stabilized lateritic soil are given below. Graphically determine the design CBR value of the soil required to design a flexible pavement if 150 divisions of the proving ring division represents 280 kg load in the calibration chart of the proving ring. Dial gauge reading. Show all the necessary calculations with relevant units. Take least count of dial gauge as 0.01 mm.

Dial Gauge Reading (Div)	Proving Ring Reading (Div)	
	Sample 1	Sample 2
0	0	0
50	35	03
100	60	04
150	84.5	16
200	102	32
250	116	48.5
300	127	65
350	135.5	81
400	143.5	96
450	151.5	110
500	158	124
550	163	135
600	167	143

10 L*5 1 3

- b) A dual wheel load assembly with 60 kN load on each wheel and contact pressure of 0.7 MPa is applied on a homogeneous mass with modulus of elasticity 12 MPa. If the center-to-center distance between the two wheels is 300mm, determine the **vertical stress** value at a depth of 0.7 m at following four points along the line joining the centers of two wheel loads:

- (i) At the center of the dual wheels
 (ii) Exactly below the centre of one of the wheel
 (iii) Mid Point of (i) and (ii).

10 L3 2 2

2. a) Describe the influence of **Design wheel load** and **Environmental Factors** in design and performance of a flexible pavement.

6+4 L2 1 1

- b) Plate load tests were conducted using 300 mm diameter plate on subgrade on a granular base course of thickness 450 mm sustained a pressure of 1.25 kg/cm^2 and 8.0 kg/cm^2 respectively at 5.0 mm deflection. Design the flexible pavement for a wheel load of 5100 kg & contact pressure of 7.0 kg/cm^2 so that to cause the deflection of 5.0 mm.
- 10 L4 2 2
3. a) Explain with a neat sketch typical component of a rigid pavement. Also list any four design features to show how this pavement is different from its counterpart.
- 10 L2 1 1
- b) Define **ESWL** by stress and deflection criteria and, determine the ESWL for the dual-tandem wheel load assembly at a depth of 800 mm and 1250 mm by graphical method using the following data:
 Load on each wheel = 70 kN
 Contact pressure = 0.85 MPa
 c/c Spacing of dual wheels = 500 mm
 c/c Spacing of tandem axles = 1000 mm
 (make necessary calculations and use plain graph sheet paper)
- 2+8 L4 2 3
- Unit – II**
4. a) A CC pavement slab of thickness 20 cm is constructed over a granular sub-base having modulus of reaction 15 kg/cm^3 . The maximum temperature difference between the top and the bottom of the slab during summer day & night is found to be 18°C . The spacing between the transverse contraction joint is 4.5m and that between longitudinal joint is 3.5m. The design wheel load is 5100kg, radius of contact area is 15cm, E value of CC is $3 \times 10^5 \text{ kg/cm}^2$, Poisson's ratio = 0.15, & coefficient of thermal expansion of CC is 10×10^{-6} per $^\circ\text{C}$ and friction coefficient is 1.5. Using the *design charts* suggested by IRC and chart for warping stress coefficient, find the worse combination of stresses.
- 10 L3 3 2
- b) With a neat sketch, explain the main components of the typical cross-section of a highway on embankment. Also, indicate any two functions of an embankment.
- 8+2 L2 4 1
5. a) A cement concrete pavement is to be designed for a two lane two way National Highway in Karnataka State for a design life of 20 years. The total two-way traffic is 3000 CVPD at the end of construction period. Use IRC:58-2002 Guidelines and charts for the design. The design Parameters are:
- Flexural Strength of Cement Concrete = 45 kg/cm^2
 - Modulus of subgrade reaction = 8 kg/cm^3
 - Elastic modulus of concrete = $3 \times 10^5 \text{ kg/cm}^2$
 - Poisson's ratio = 0.15
 - Thermal coefficient of concrete = $10 \times 10^{-6} / ^\circ\text{C}$
 - Tyre pressure = 8 kg/cm^2
 - Traffic growth rate = 7.5%
 - Spacing of Contraction joints = 4.5m
 - Width of the pavement = 3.5 m
 - 98 percentile axle load = 16 tonnes.
- The axle load spectrum obtained from axle load survey is given in the following table:

Single Axle Load		Tandem Axle Load	
Axle Load Range (tonnes)	Percentage of axle load	Axle Load Range (tonnes)	Percentage of axle load
19-21	0.6	34-38	0.3
17-19	1.5	30-34	0.3
15-17	4.8	26-30	0.6
13-15	10.8	22-26	1.8
11-13	22	18-22	1.5
9-11	23.3	14-18	0.5
7 - 9	30	10-14	2

- b) List the joints that are provided in CC pavement? Explain with neat sketches the construction of expansion and contraction joints?

10 L4 3 3
2+8 L2 4 .3

6. a) The design thickness of a CC pavement is 26 cm considering a design axle load of 12,000 kg on single axle and M-40 concrete. The radius of relative stiffness is found to be 62.2 cm. If the elastic modulus of dowel bar steel is 2×10^6 kg/cm², the modulus of dowel-concrete interaction is 41,500 kg/cm³ and joint width is 1.8 cm, design the dowel bars for 40 % load transfer considering edge loading.

10 L3 3 2

- b) Explain the specifications of materials, construction method and field control checks for Cement Concrete Pavement Construction as per MORTH guidelines.

3+3+4 L3 4 2

Unit – III

7. a) While conducting BBD studies, the initial dial gauge reading, intermediate reading, and final reading obtained at two deflection observation points A and B are given below. If the least count of the dial gauge is 0.01 mm, and the value of K of Benkelman beam is 2.91; Determine the rebound deflection values at the two points.

	D _o	D _i	D _f
Point A	0	46	44
Point B	0	33	29
Point C	05	60	59

10 L3 5 3

- b) Explain the step-by step procedure for construction of WBM layer as per MORTH guidelines. Indicate the quality control checks during the construction of the same.

6+4 L2 5 1

8. a) Write a note on General approach and Field data collection during BBD studies as per IRC 81 guidelines.

10 L2 5 1

- b) Explain the step-by step procedure for construction of DBM sub-surface layer as per MORTH guidelines. Indicate the quality control checks during the construction of the same.

6+4 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
